

# Cristoff Stan

## Senior Backend Engineer - Rust, AWS

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### PROFESSIONAL SUMMARY

Senior Backend Engineer with 10+ years delivering B2B SaaS, e-commerce, and enterprise operations platforms using Rust, Cargo, Tokio, Axum, Actix Web, Rocket, Serde, SQLx, Diesel, async/await, ownership, lifetimes; combines backend and distributed-systems architecture, legacy modernization, cloud-native delivery, performance engineering, and technical leadership to build secure, scalable systems that improve reliability, release velocity, operational efficiency, and measurable business outcomes.

### TECHNICAL SKILLS

- **Backend Engineering:** Rust, Cargo, Tokio, Axum, Actix Web, Rocket, Serde, SQLx, Diesel, async/await, ownership, lifetimes, traits, channels and memory-safe concurrency.
- **API and Distributed Systems:** REST APIs, GraphQL, gRPC, WebSockets, background workers, schedulers, streaming services, pagination, rate limiting, idempotency, API versioning and resilient integrations.
- **Architecture and Design:** Microservices, modular monoliths, Domain-Driven Design, clean architecture, hexagonal architecture, CQRS, event-driven architecture, API gateways, OAuth 2.0, OpenID Connect, JWT, RBAC, multi-tenancy, transactional outbox, saga workflows, idempotency, rate limiting and secure API design.
- **Data and Messaging:** PostgreSQL, MySQL, MongoDB, Redis, Kafka, NATS and cloud messaging, SQL optimization, indexing, connection pooling, schema migrations and distributed caching.
- **Cloud Platforms:** AWS Lambda, ECS, EKS, EC2, API Gateway, RDS, DynamoDB, S3, CloudFront, SQS, SNS, EventBridge, IAM, KMS and CloudWatch.
- **DevOps and Infrastructure:** Docker, Kubernetes, Terraform, Helm, GitHub Actions, GitLab CI, Jenkins, Argo CD, infrastructure as code, automated quality gates, blue-green deployments, canary releases, rollback planning, container scanning and environment promotion.
- **Testing and Quality:** Cargo test, property-based testing, mockall, criterion, cargo-fuzz, clippy, rustfmt, Testcontainers, Pact and k6.
- **Observability and Leadership:** OpenTelemetry, Prometheus, Grafana, Datadog, New Relic, Sentry, Elastic Stack, structured logging, distributed tracing, SLI/SLO monitoring, alert tuning, incident response and root-cause analysis. Architecture reviews, technical leadership, mentoring, code reviews, legacy modernization, performance engineering, production readiness, secure development, Agile delivery, technical estimation, stakeholder communication and cross-team release coordination.

### PROFESSIONAL EXPERIENCE

#### BEVY

Senior Software Engineer

September 2021 - June 2026

Remote, Romania

- Architected **multi-tenant B2B SaaS and enterprise operations platforms** using **Rust 1.45-1.85, Tokio, Axum/Actix, and SQLx, PostgreSQL/Redis, event messaging, and AWS**, defining scalable domain boundaries, reusable delivery standards, and production controls across several product teams.
- Modernized tightly coupled legacy applications into **modular, cloud-ready services and interfaces** aligned with **Rust, Cargo, Tokio, Axum, Actix Web, Rocket, Serde, SQLx, Diesel, async/await, ownership, lifetimes, traits, channels and memory-safe concurrency**, using staged migration, compatibility APIs, feature flags, and contract validation to protect active customers.
- Reduced **p95 API latency by 41%** by eliminating inefficient queries, adding composite indexes, introducing pagination and caching, and moving repeated calculations into optimized services.
- Resolved production incidents involving **database-pool exhaustion, memory growth, duplicate background jobs, stale cache entries, partial downstream failures, and unbounded retries** by adding lifecycle controls, idempotency, correlation IDs, and failure isolation.
- Led safe upgrades across **Rust 1.45-1.85, Tokio, Axum/Actix, and SQLx, PostgreSQL/Redis, event messaging, and AWS** by replacing unsupported dependencies, updating build images and deployment manifests, validating breaking changes through regression and load testing, and releasing through canary stages with rollback plans.
- Delivered new capabilities for **billing, subscriptions, reporting, audit history, notifications, search, administration, and customer-support workflows**, translating complex requirements into secure APIs, resilient asynchronous processing, and maintainable user experiences.
- Improved architecture with **Domain-Driven Design, clean architecture, event-driven workflows, transactional outbox processing, tenant-aware authorization, distributed caching, and observability**, strengthening consistency across synchronous and asynchronous operations.

- Strengthened **CI/CD and production quality** with **Docker, Kubernetes, Terraform, automated tests, security scanning, schema-migration gates, deployment approvals, and AWS monitoring**, reducing release risk and improving operational visibility.
- Mentored engineers through **architecture reviews, pull-request feedback, production debugging, incident retrospectives, estimation, and release-readiness sessions**, improving code consistency, technical ownership, and cross-team delivery decisions.

## Syscom Digital

October 2019 - August 2021

Software Engineer

Remote, Romania

- Developed commerce, customer-service, and operational products using **Rust 1.29-1.56, async services, and PostgreSQL integration, PostgreSQL/Redis, event messaging, and AWS**, supporting checkout, order tracking, refunds, fulfillment, reporting, user administration, and third-party provider integrations.
- Implemented new **payment, notification, search, workflow, export, and reconciliation features** with secure APIs, asynchronous workers, validation, and audit-friendly state transitions across customer and administrative experiences.
- Reduced **payment and webhook failures by 31%** through idempotency keys, signature validation, correlation identifiers, bounded retries, dead-letter handling, and reconciliation jobs.
- Investigated production issues involving **duplicate provider callbacks, inventory mismatches, stale sessions, timeout chains, malformed payloads, and long-running database transactions**, then introduced concurrency controls, bounded retries, and reconciliation routines.
- Modernized older modules within **Rust 1.29-1.56, async services, and PostgreSQL integration, PostgreSQL/Redis, event messaging, and AWS** by introducing typed contracts, clearer service boundaries, shared components, supported runtime versions, and compatibility tests that enabled gradual replacement without interrupting daily operations.
- Improved performance through **server-side pagination, indexed queries, reduced serialization, selective data loading, caching, connection-pool tuning, and browser or runtime profiling**, keeping high-volume administrative workflows responsive.
- Expanded delivery pipelines with **Jenkins/GitLab CI, Docker, unit and integration tests, static analysis, artifact versioning, migration validation, smoke tests, and staged environment promotion** across development, QA, and production.
- Collaborated with product owners, designers, QA, support teams, security reviewers, and external providers to convert ambiguous requirements into **testable stories, resilient failure handling, measurable acceptance criteria, and release documentation**.

## Soft Build

June 2016 - July 2019

Software Developer

Remote, Romania

- Built full-stack business applications using **Rust 1.0 and later stable releases for systems utilities, PostgreSQL/Redis, event messaging, and AWS** for commerce, booking, pricing, inventory, reporting, account management, and internal administration across multiple client engagements.
- Improved **application response time by 38%** by reducing oversized payloads, optimizing database access, caching stable reference data, and moving blocking work into background processing.
- Developed new **REST endpoints, reusable interface modules, scheduled imports, reporting features, role-based workflows, and provider integrations**, keeping validation and business rules separated from delivery and persistence concerns.
- Resolved defects involving **duplicate submissions, stale inventory, null-reference failures, inconsistent validation, slow database queries, missing error feedback, and partial updates** through transactions, defensive checks, indexed access, and explicit recovery states.
- Moved blocking email, document, synchronization, import, and report-generation work into **background jobs and queue-based processing**, adding retry policies, schedules, failed-job diagnostics, and operational visibility for support teams.
- Expanded quality coverage with **unit, integration, API regression, browser, and database migration tests**, prioritizing authentication, pricing, persistence, concurrency, and other high-risk customer workflows.
- Supported **execution-plan analysis, production incident resolution, deployment automation, code reviews, technical estimation, and legacy modernization**, while introducing reusable architecture and coding standards across projects.

## Decima Digital

November 2015 - April 2016

Software Developer

Remote, Estonia

- Delivered customer-facing websites and administration tools using **memory-safe systems utilities and C-compatible integrations, PostgreSQL/Redis, event messaging, and AWS**, supporting account workflows, content management, file handling, search, forms, notifications, and operational reporting.
- Maintained server-rendered and tightly coupled legacy modules while introducing **reusable services, clearer controllers, modular client code, and consistent validation**, allowing new features to coexist safely with older functionality.
- Built controllers, service classes, repositories, scheduled utilities, and database scripts for **content retrieval, form processing, account operations, file uploads, notification routing, and report generation**.

- Fixed production defects involving **failed uploads, broken validation, malformed requests, null-reference exceptions, duplicate submissions, database connectivity errors, and missing user feedback** through reproducible diagnostics and defensive handling.
- Improved slow operations by reviewing generated SQL, removing unnecessary object loading, adding targeted indexes, reducing repeated in-memory loops, and replacing oversized queries with focused data access patterns.
- Participated in **deployment preparation, browser and API compatibility testing, backup verification, smoke testing, rollback planning, and requirement reviews** with designers, project managers, and senior engineers.

## Borna Technology

September 2014 - August 2015

Software Developer Intern

Tallinn, Estonia

- Assisted senior engineers with **systems programming fundamentals and low-level debugging, PostgreSQL/Redis, event messaging, and AWS** across internal tools and client-facing applications, following established coding, source-control, database, and release practices.
- Built small interface enhancements, reusable templates, server-side handlers, validation scripts, and database-backed administration pages while learning structured debugging and maintainable implementation patterns.
- Created and updated **SQL queries, stored procedures, migration scripts, test data, and integration utilities** under supervision, applying safe database-change and review practices.
- Investigated **invalid submissions, null-reference errors, database connectivity failures, layout defects, inconsistent API responses, and deployment configuration problems** by reproducing issues and documenting diagnostic steps.
- Contributed to **QA checks, deployment notes, smoke tests, code reviews, and technical documentation**, establishing a practical foundation in collaborative delivery, production support, and release discipline.

## SELECTED PROJECTS

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### Cloud-Native Commerce and Operations Platform

Designed a multi-tenant platform using Rust, Cargo, Tokio, Axum, Actix Web, Rocket, Serde, SQLx, Diesel, async/await, ownership, lifetimes, traits, channels and memory-safe concurrency to support order management, payment reconciliation, inventory, customer support, reporting, audit history and event-driven notifications.

Implemented idempotent APIs, transactional outbox processing, tenant-aware authorization, distributed tracing, integration tests and containerized CI/CD pipelines.

### Enterprise Application Modernization Program

Led the migration of a legacy application into services and interfaces built with Rust, Cargo, Tokio, Axum, Actix Web, Rocket, Serde, SQLx, Diesel, async/await, ownership, lifetimes, traits, channels and memory-safe concurrency, replacing tightly coupled modules with documented APIs and independently deployable workloads.

Resolved outdated dependencies, inefficient queries, inconsistent authentication, fragile schema changes and limited production visibility through staged modernization and automated testing.

## EDUCATION

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Tallinn University - Bachelor's Degree in Computer Science | September 2011 - May 2014